Reg. No.										

MANIPAL INSTITUTE OF TECHNOLOGY

MANIPAL

(A constituent unit of MAHE, Manipal)

SEVENTH SEMESTER BTECH. (E & C) DEGREE END SEMESTER EXAMINATION
DECEMBER 2021-JANUARY 2022
SUBJECT: RADAR AND NAVIGATION SYSTEMS (ECE -4084)

TIME: 75 minutes **MAX. MARKS: 20**

Instructions to candidates

- Answer **ALL** questions.
- Missing data may be suitably assumed.

Q. No.	Questions	Marks
1A.	Starting from fundamentals, derive an expression for a radar range, while considering the receiver noise, and discuss the factors affecting the maximum range of a radar	5
1B.	A radar transmitter has a peak pulse power of 300kW , a PRF of 1200pps and a pulse width of 0.9µS. Calculate (i) the maximum unambiguous range (ii) the duty cycle (iii) the average transmitted power	3
1C.	A Doppler radar operating at 12GHz, is intended to detect target velocities ranging from 14.4km/hr to 108km/hr. What is the required passband of the Doppler filter?	2
2A.	With neat diagrams, explain instrument landing system used in aircraft landing. Compare it with GAGAN system	5
2B	An earth station with a transmitter power of 120W, a frequency of 6GHz, and an antenna gain of 42dB transmits to a satellite repeater. The receiver antenna on the satellite has a gain of 31 dB, and the satellite is in a synchronous orbit 35900 km above the earth. What is the received power, in dBm?	3
2C	An FM CW Radar working at 4GHz has frequency excursions of 60MHz, a modulating frequency of 100Hz, and a range of 300 meters. Find the beat note frequency	2